

Attention Is All You Need

Ashish Vaswani*, Noam Shazeer*, Niki Parmar*, Jakob Uszkoreit*,
Llion Jones*, Aidan N. Gomez*, Lukasz Kaiser*, Illia Polosukhin*

Google Brain, Google Research, University of Toronto

Abstract

This fixture reproduces only the bibliographic front matter of a well known preprint; the abstract and all body paragraphs are placeholder text written for testing. It is used to check that titles, authors and identifiers are recovered from the printed page when the document information dictionary is wrong.

1 Introduction

This document is a test fixture for SoloPDF. It imitates the front page of a machine learning preprint so that the citation extractor has something realistic to read: a large title, a list of authors with affiliation marks, an abstract and two columns of running text.

The paragraphs below are filler. They exist so that the reading ruler has many lines to walk through, and so that those lines sit in two columns whose baselines do not quite agree. A reader moving down the left column should stay in the left column until it ends, and only then continue at the top of the right one.

Line focus tools are an accessibility aid. People with dyslexia or attention difficulties often lose their place on a dense page; a band that brightens the current line and dims the rest keeps the eye anchored without changing the layout of the document itself.

2 Background

Desktop readers have offered a magnifying loupe for many years. Holding a key shows a circle that renders the area under the pointer at a higher resolution, which helps with small print in figures, footnotes and scanned tables.

Reference managers made a different habit popular: copying a formatted citation straight from the document being read. The metadata is often incomplete, so a good reader combines the document information dictionary, embedded XMP and what is printed on the first page.

3 Method

We read the first two pages, group text runs into lines, and treat the largest type near the top of the first page as the title. Name-shaped lines directly underneath become the author list, and identifiers such as a DOI or an arXiv number are matched with conservative patterns.

Online enrichment is never automatic. Only when the reader asks for exact metadata does the application contact the DOI resolver, which answers with structured data in the citation style language format.

4 Evaluation

The fixture deliberately ships a useless information dictionary: its title is the name of a word processor file and its author is an account name. Both must be ignored in favour of the printed front matter.

The second page cites another work by its DOI. That identifier belongs to the cited work, not to this document, and must not be reported as the DOI of this paper.

Further filler text follows to fill the right column. Reading aids are only useful when they are predictable, so the ruler snaps to real text lines rather than to fixed pixel steps, and it follows the reading order of columns.

When the last line on a page is reached, moving further continues on the next page. In paged layouts this turns the page; in continuous layouts the view scrolls so that the band stays comfortably inside the window.

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References

- [1] A. Author and B. Author. Reading documents at scale. *Journal of Tools*, 12(3):1-20, 2019. doi:10.1000/other.2019.001
- [2] C. Writer. A survey of PDF readers. Tech. report, 2020.